Egg segmentation and tracking

Computer Vision Intern Task | reproducible experimental report

Executive summary

A pretrained MobileSAM model produced reference egg masks for all 110 images. A classical local graph-cut method, selected on the development set, achieved mean IoU **0.565** and Dice **0.648** on 22 evaluation images. A two-ID tracker processed the complete 60-second reference video and marks uncertain reidentification after occlusion.

Dataset and split

Roboflow egg v1 classification export: 110 images (108 in the egg folder and 2 Unlabeled), 640 x 640, CC BY 4.0. The supplied ZIP is v1; the online v2/v3 pages list 160 images but were not the provided local export. A deterministic 4:1 split gives 88 development and 22 evaluation images. pHash and geometrically verified ORB matches form 103 scene groups; all members of a group remain on one side of the split.



The provided images span hands, ash, cookware, fabric and patterned backgrounds.

Task 1 - pretrained reference segmentation

MobileSAM automatic proposals at 384 px were filtered by area, shape, border contact and overlap. Visual audit found missed eggs, so 23 images received human-selected positive prompts; four difficult images used boxes. Prompts were chosen from the image, independent of classical predictions. Two boiling-water images have no visible egg mask. All masks, prompt coordinates and model checksum are saved. These are **pseudo-labels**, not human ground truth; agreement with them does not prove absolute accuracy.

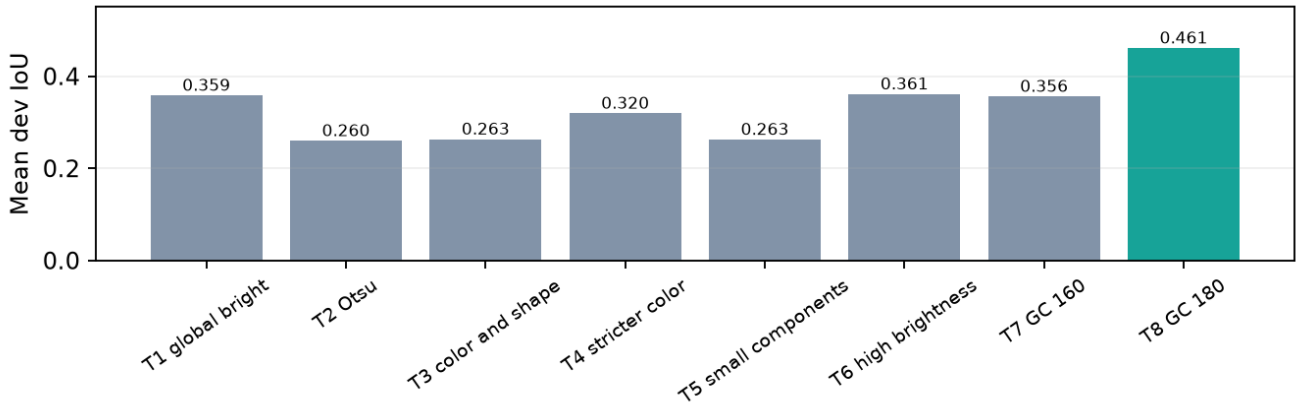
Task 2 - classical method

The final method converts each frame to grayscale and HSV, selects bright low-saturation seed regions, removes tiny noise, filters implausible components by shape, and runs local GrabCut around each seed. GrabCut fits foreground/background color models and expands a seed toward the full egg boundary. The output is a union of candidate masks. The video uses a documented stricter brightness setting and table ROI; it does not use MobileSAM at runtime.

Experiment log

Each trial used the same 88-image development set. Parameters and reasons are in outputs/experiment_log.json; the table below gives the decision path. The selection rule was highest mean dev IoU.

Trial	Change / reason	Dev IoU	Decision
T1 global bright	Simple light-egg baseline	0.359	Drop
T2 Otsu	Adaptive global threshold for varying exposure	0.260	Drop
T3 color and shape	Reject colored background and non-elliptical regions	0.263	Drop
T4 stricter color	Reduce bright clutter false positives	0.320	Drop
T5 small components	Recover distant eggs	0.263	Drop
T6 high brightness	Test stronger foreground contrast	0.361	Drop
T7 graph cut 160	Grow bright egg cores to full boundaries using local color models	0.356	Drop
T8 graph cut 180	Stricter seeds before local graph cut	0.461	Keep

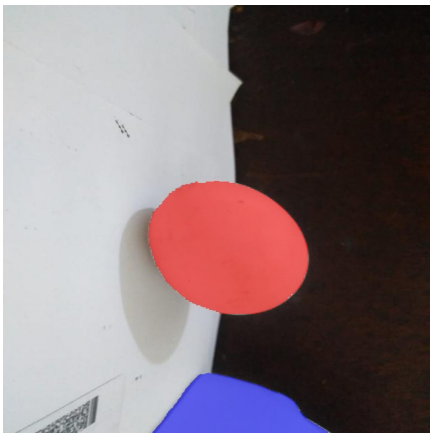


Task 3 - held-out evaluation

Mean IoU 0.565 (image-bootstrap 95% interval 0.416-0.705); Dice 0.648; pixel precision 0.712; pixel recall 0.677. Object matching at IoU ≥ 0.5 yields precision 0.644 and recall 0.600; mean absolute count error 1.00. Metrics are per-image macro means; the bootstrap interval reflects small-sample variation, not label uncertainty.

Qualitative error analysis

Green = overlap; red = MobileSAM only; blue = classical method only. Samples are the lowest-IoU evaluation cases, chosen by a fixed ranking. Open outputs/explorer/index.html for the full interactive gallery.



not_damaged_30_jpg
IoU 0.000



not_damaged_135_jpg
IoU 0.000



not_damaged_150_jpg
IoU 0.000

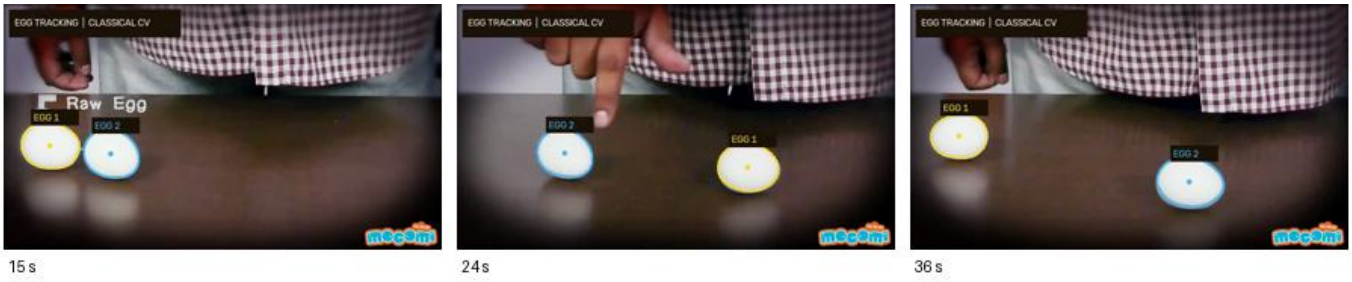
The first case selects bright tissue while missing the egg. The second is an Unlabeled boiling-water frame: a reflection is falsely predicted as an egg. The third misses a shadowed egg entirely. These cases expose different failure mechanisms rather than one generic threshold error.

Observed gaps

Dark or shadowed eggs can fail the bright-seed step. Bright cookware, tissue and reflections can become false positives. GrabCut can absorb a similar-colored background or split an egg occluded by a rail. The reference pipeline itself misses partially submerged eggs and sometimes selects non-egg proposals; the saved QC sheet and manual prompt log make this visible. Next improvements: human instance masks for a representative subset, illumination normalization, temporal consistency for video, and a dedicated egg detector to provide proposals.

Video tracking and reproducibility

Task 4 uses a table ROI and a stricter brightness seed to suppress hands, captions and reflections. Connected components are assigned to two persistent tracks with global minimum-distance matching; a frame gap over 24 frames or a jump over 100 px is marked uncertain. Across 1,440 video frames, the output contains 1351 egg detections and 3 uncertain reidentification events. Identity during hand occlusion cannot be established confidently from two visually similar eggs; these events are explicitly signalled rather than silently relabelled.



Three frames from the annotated output. IDs remain distinct; an uncertain recovery is marked with ?.

Tracking video

Public Google Drive video: https://drive.google.com/file/d/1oLbEB3EDDzFkoCDZdwqK_lTCbqnHanMv/view?usp=drivesdk

Reproduce

Create a Python 3.12 environment, install requirements.txt, keep the supplied egg.v1i.folder at the repository root, and run scripts/generate_reference.py, scripts/repair_reference.py, scripts/run_experiments.py, scripts/track_video.py, and scripts/make_report.py in that order. See README.md for exact commands. The MobileSAM checkpoint is downloaded separately and identified by checksum in outputs/reference_provenance.json.

Attribution and AI use

Dataset: Fire, egg v1, Roboflow Universe, CC BY 4.0, https://universe.roboflow.com/fire-bf7qe/egg-vzhtr/dataset/1 . Model: Ultralytics MobileSAM, https://docs.ultralytics.com/models/sam/ . Source video: Mocomi, https://www.youtube.com/watch?v=Y4JyV94y2_A . AI coding assistance was used for code drafting, debugging, and report wording; the workflow, visual QC, method comparison, and interpretation are documented for independent review.

Limitations

The dataset contains only classification folders, not segmentation annotations. The same images were inspected during development, so the reported agreement is exploratory and should be confirmed with independent human-labelled masks. Scene grouping reduces but cannot eliminate leakage from visually related images. The video-specific ROI is tailored to this clip and should not be assumed to transfer to other scenes.